Pareenita Atul Shirsath BE AI&DS Roll.no: 49

SMA EXPERIMENT NO : 04

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from google.colab import files
import os
from textblob import TextBlob

# Load data
file_name = 'Youtube-dataset-sample.csv'
if not os.path.exists(file_name):
    print(f"The file '{file_name}' was not found. Please upload it.")
    uploaded = files.upload()
    if file_name not in uploaded:
        raise FileNotFoundError(f"'{file_name}' was not uploaded. Please ensure you upload the correct file.")
df = pd.read_csv(file_name)

# Sentiment analysis using TextBlob
def analyze_sentiment(text):
    if pd.isna(text):
        return 'Neutral'

    polarity = TextBlob(str(text)).sentiment.polarity

    if polarity > 0:
        return 'Positive'
    elif polarity < 0:
        return 'Negative'
    else:
        return 'Neutral'

# Apply sentiment analysis
df['sentiment'] = df['comment_text'].apply(analyze_sentiment)

# Results
print("Sentiment Distribution:")
print(df['sentiment'].value_counts())
print("\nPercentages:")
print(df['sentiment'].value_counts(normalize=True) * 100)

# Create 5 visualizations
fig, axes = plt.subplots(2, 3, figsize=(18, 10))
colors = ['green', 'gray', 'red']

# 1. Bar Chart
df['sentiment'].value_counts().plot(kind='bar', ax=axes[0,0], color=colors)
axes[0,0].set_title('1. Bar Chart - Sentiment Distribution')
axes[0,0].set_ylabel('Count')
axes[0,0].set_xticklabels(axes[0,0].get_xticklabels(), rotation=0)

# 2. Pie Chart
df['sentiment'].value_counts().plot(kind='pie', ax=axes[0,1], autopct='%1.1f%%', colors=colors)
axes[0,1].set_title('2. Pie Chart - Sentiment Percentages')
axes[0,1].set_ylabel('')

# 3. Box Plot - Sentiment vs Likes
df.boxplot(column='likes', by='sentiment', ax=axes[0,2])
axes[0,2].set_title('3. Box Plot - Likes by Sentiment')
axes[0,2].set_xlabel('Sentiment')
axes[0,2].set_ylabel('Likes')
plt.suptitle('')

# 4. Heatmap - Sentiment vs Replies
df['replies_cat'] = pd.cut(df['replies'], bins=[0, 1, 3, 10], labels=['Low', 'Medium', 'High'])
heatmap_data = pd.crosstab(df['sentiment'], df['replies_cat'])
sns.heatmap(heatmap_data, annot=True, fmt='d', cmap='Blues', ax=axes[1,0])
axes[1,0].set_title('4. Heatmap - Sentiment vs Replies')

# 5. Scatter Plot - Likes vs Replies
for sentiment, color in zip(['Positive', 'Neutral', 'Negative'], colors):
    mask = df['sentiment'] == sentiment
    axes[1,1].scatter(df[mask]['likes'], df[mask]['replies'],
```

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                    c=color, label=sentiment, alpha=0.6, s=50)

axes[1,1].set_title('5. Scatter Plot - Likes vs Replies')
axes[1,1].set_xlabel('Likes')
axes[1,1].set_ylabel('Replies')
axes[1,1].legend()
axes[1,1].grid(alpha=0.3)

# Hide empty subplot
axes[1,2].axis('off')

plt.tight_layout()
plt.show()
```

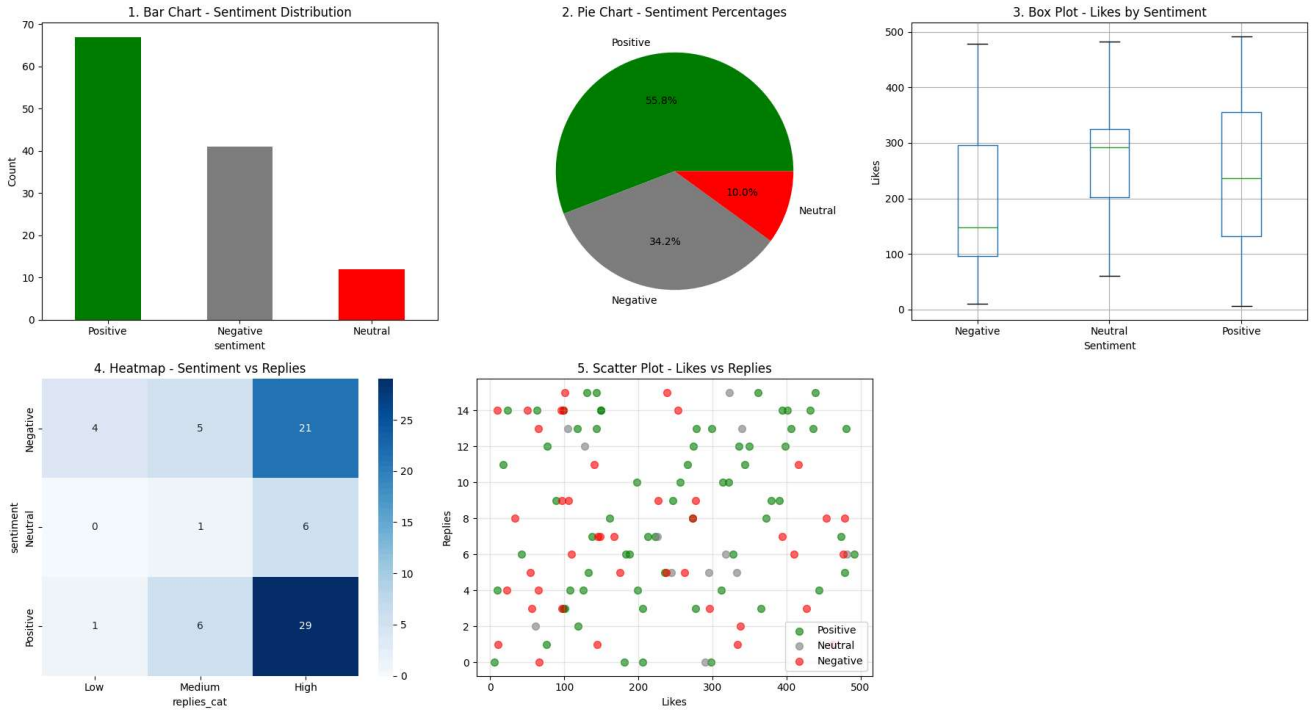
```
The file 'Youtube-dataset-sample.csv' was not found. Please upload it.
```

Choose Files Youtube-da…-sample.csv

Youtube-dataset-sample.csv(text/csv) - 3139 bytes, last modified: 15/3/2026 - 100% done

```
Saving Youtube-dataset-sample.csv to Youtube-dataset-sample.csv
Sentiment Distribution:
sentiment
Positive    67
Negative    41
Neutral     12
Name: count, dtype: int64

Percentages:
sentiment
Positive    55.833333
Negative    34.166667
Neutral     10.000000
Name: proportion, dtype: float64
```




```
!pip install google-api-python-client textblob seaborn
```

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Requirement already satisfied: textblob in /usr/local/lib/python3.12/dist-packages (0.19.0)
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib!=3.6.1
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.20.0
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.20.0->google-api
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.20.0->goog
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.20.0->goog
```

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from googleapiclient.discovery import build
from textblob import TextBlob
```

```
API_KEY = "AIzaSyDFZViTcfxxjFMr67zLyr82Z7ylElnipKg"

youtube = build(
    "youtube",
    "v3",
    developerKey=API_KEY
)
```

```
video_id = "https://www.youtube.com/watch?v=xPh5ihBWang"
```

```
import urllib.parse

comments = []
likes = []
replies = []

# Extract video ID from the URL
query_params = urllib.parse.parse_qs(urllib.parse.urlparse(video_id).query)
actual_video_id = query_params.get('v', [video_id])[0]

request = youtube.commentThreads().list(
    part="snippet",
    videoId=actual_video_id,  # Use the extracted video ID
    maxResults=100,
    textFormat="plainText"
)

response = request.execute()

for item in response['items']:
    comment = item['snippet']['topLevelComment']['snippet']['textDisplay']
    like_count = item['snippet']['topLevelComment']['snippet']['likeCount']
    reply_count = item['snippet']['totalReplyCount']

    comments.append(comment)
    likes.append(like_count)
    replies.append(reply_count)
```

```
df = pd.DataFrame({
    "comment_text": comments,
    "likes": likes,
    "replies": replies
})

print(df.head())
```

```
                                        comment_text  likes  replies
0                       where we can find the data,      8        0
1                                         Wowwwww😱😱       2        0
2  This course brilliantly clarifies Analyst vs. ...     23        0
```

```
def analyze_sentiment(text):
    polarity = TextBlob(text).sentiment.polarity

    if polarity > 0:
        return "Positive"
    elif polarity < 0:
        return "Negative"
    else:
        return "Neutral"

df["sentiment"] = df["comment_text"].apply(analyze_sentiment)
```

```
print("Sentiment Distribution:")
print(df["sentiment"].value_counts())

print("\nPercentages:")
print(df["sentiment"].value_counts(normalize=True)*100)
```

```
Sentiment Distribution:
sentiment
Neutral     2
Positive    1
Name: count, dtype: int64

Percentages:
sentiment
Neutral     66.666667
Positive    33.333333
Name: proportion, dtype: float64
```

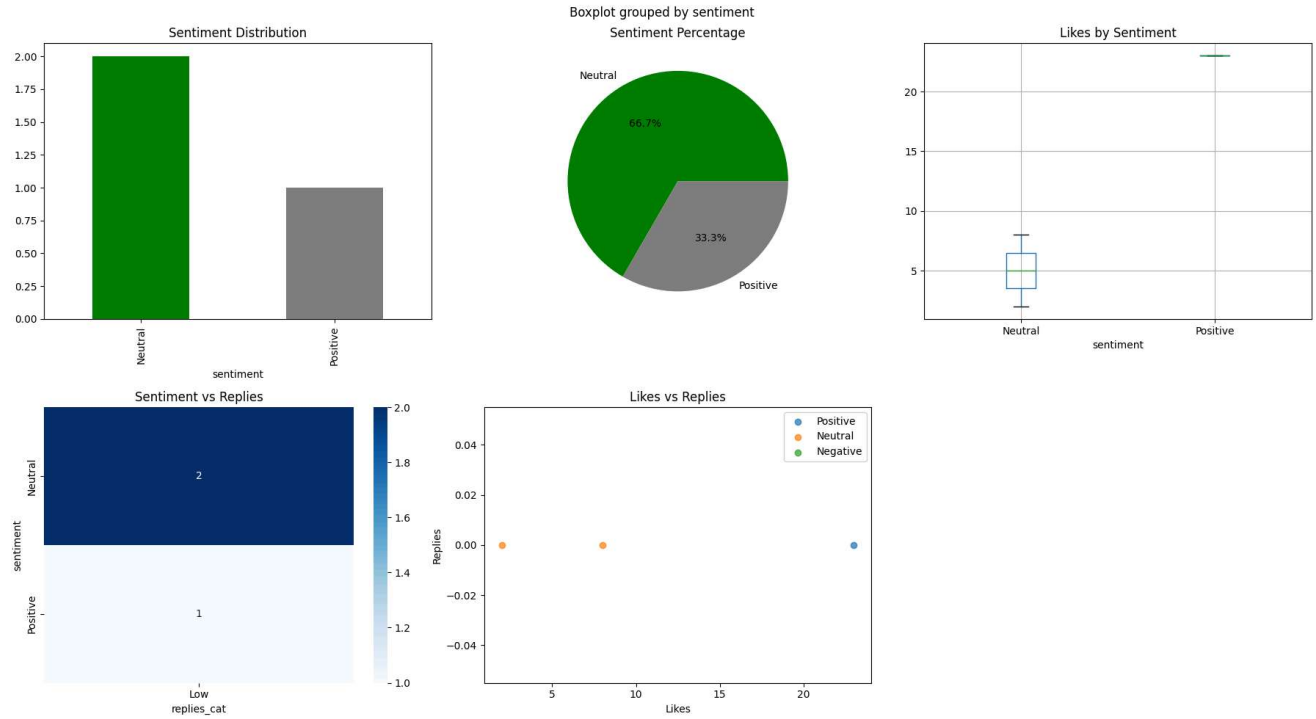
```
fig, axes = plt.subplots(2,3, figsize=(18,10))

colors = ["green","gray","red"]

# 1 Bar Chart
df['sentiment'].value_counts().plot(
    kind="bar",
    ax=axes[0,0],
    color=colors
)
axes[0,0].set_title("Sentiment Distribution")

# 2 Pie Chart
df['sentiment'].value_counts().plot(
    kind="pie",
    autopct='%1.1f%%',
    ax=axes[0,1],
    colors=colors
)
axes[0,1].set_ylabel("")
axes[0,1].set_title("Sentiment Percentage")

# 3 Box Plot
df.boxplot(
    column="likes",
    by="sentiment",
    ax=axes[0,2]
)
axes[0,2].set_title("Likes by Sentiment")

# 4 Heatmap
df["replies_cat"] = pd.cut(df["replies"], bins=[0,1,5,20],
labels=["Low","Medium","High"], include_lowest=True)
```

```
heatmap_data = pd.crosstab(df["sentiment"],df["replies_cat"])

sns.heatmap(
    heatmap_data,
    annot=True,
    cmap="Blues",
    ax=axes[1,0]
)

axes[1,0].set_title("Sentiment vs Replies")

# 5 Scatter Plot
for sentiment,color in zip(["Positive","Neutral","Negative"],colors):

    mask = df["sentiment"]==sentiment

    axes[1,1].scatter(
        df[mask]["likes"],
        df[mask]["replies"],
        label=sentiment,
        alpha=0.7
    )

axes[1,1].set_title("Likes vs Replies")
axes[1,1].set_xlabel("Likes")
axes[1,1].set_ylabel("Replies")
axes[1,1].legend()

axes[1,2].axis("off")

plt.tight_layout()
plt.show()
```